第 5 章

Higher condensation theory

Review of [?].

5.1 Definition of condensation

In this section, we refer to an **(n, n)-category** as an (∞, n)-category which is "truncated" at morphism degree n. Intuitively, if $\mathcal{C}$ is an (n, n)-category, then, for arbitrary $n-1$-morphisms $f, g \in \mathcal{C}_{n-1}$, the space of n-morphisms between f and g has discrete topology. Therefore, we can compose n-morphisms strictly (not up to homotopy), associatively, and unitally.

Note that a **$(0, 0)$-category** $\mathcal{C}$ is merely a set[*1].

定義 5.1: 0-condensation

Let $\mathcal{C}$ be a $(0, 0)$-category. A **0-condensation** on $\mathcal{C}$ is an equality between elements of a set $\mathcal{C}_0$.

n-condensation is defined by induction.

定義 5.2: n-condensation

Fix $n \geq 0$. Let $\mathcal{C}$ be an (n, n)-category, and let $x, y \in \mathcal{C}_0$ be objects of $\mathcal{C}$. We define n-condensation by induction on n.

n-condensation of x onto y in $\mathcal{C}$ consists of three data:

- A 1-morphism $r \in \mathrm{Map}_{\mathcal{C}}(x, y)_0$
- A 1-morphism $i \in \mathrm{Map}_{\mathcal{C}}(y, x)_0$
- An $(n-1)$-condensation of $r \circ i$ onto Id_y in $\mathrm{Map}_{\mathcal{C}}(y, y)$[a]

[a] Roughly speaking, a mapping space $\mathrm{Map}_{\mathcal{C}}(y, y)$ itself is an $(n-1, n-1)$-category. Strickly speaking, we need enriched ∞-category theory.

After [?], we denote an n-condensation of x onto y in $\mathcal{C}$ as $x \longrightarrow y$.

[*1] That is, $\mathcal{C}$ consists of a set $\mathcal{C}_0$ of objects (0-morphisms) only.

【例 5.1.1】1-condensation

Let $\mathcal{C}$ be a (1, 1)-category (i.e. an ordinary category). **1-condensation** of x onto y in $\mathcal{C}$ consists of these data:

- A 1-morphism $r \in \mathrm{Hom}_{\mathcal{C}}(x, y)$
- A 1-morphism $i \in \mathrm{Hom}_{\mathcal{C}}(y, x)$
- A 0-condensation of $r \circ i$ onto Id_y in a (0, 0)-category $\mathrm{Hom}_{\mathcal{C}}(y, y)$. i.e. $r \circ i = \mathrm{Id}_y$.

In the context of (1, 1)-categories, such y is called a **retract of x**.

【例 5.1.2】2-condensation

Let $\mathcal{C}$ be a (2, 2)-category (i.e. a bicategory). **2-condensation** of x onto y in $\mathcal{C}$ consists of these data:

- A 1-morphism $r \in \mathrm{Map}_{\mathcal{C}}(x, y)_0$
- A 1-morphism $i \in \mathrm{Map}_{\mathcal{C}}(y, x)_0$
- A 1-condensation of $r \circ i$ onto Id_y in a (1, 1)-category $\mathrm{Map}_{\mathcal{C}}(y, y)$.

By【例 5.1.1】,

- A 1-morphism $r \in \mathrm{Map}_{\mathcal{C}}(x, y)_0$
- A 1-morphism $i \in \mathrm{Map}_{\mathcal{C}}(y, x)_0$
- A 2-morphism (in $\mathcal{C}$) $(r \circ i \xrightarrow{\rho} \mathrm{Id}_y) \in \mathrm{Map}_{\mathcal{C}}(y, y)_1$
- A 2-morphism (in $\mathcal{C}$) $(\mathrm{Id}_y \xrightarrow{\iota} r \circ i) \in \mathrm{Map}_{\mathcal{C}}(y, y)_1$
- An equality $\rho \circ \iota = \mathrm{Id}_{\mathrm{Id}_y}$

【例 5.1.3】3-condensation

Let $\mathcal{C}$ be a (3, 3)-category. By【例 5.1.2】, **3-condensation** of x onto y in $\mathcal{C}$ consists of these data:

- A 1-morphism $r \in \mathrm{Map}_{\mathcal{C}}(x, y)_0$
- A 1-morphism $i \in \mathrm{Map}_{\mathcal{C}}(y, x)_0$
- A 2-morphism $(r \circ i \xrightarrow{\rho} \mathrm{Id}_y) \in \mathrm{Map}_{\mathcal{C}}(y, y)_1$
- A 2-morphism $(\mathrm{Id}_y \xrightarrow{\iota} r \circ i) \in \mathrm{Map}_{\mathcal{C}}(y, y)_1$
- A 3-morphism

$$\left(\begin{tikzcd} \mathrm{Id}_y \arrow[r, bend left, "\rho\circ\iota"] \arrow[r, bend right, "\mathrm{Id}_{\mathrm{Id}_y}"'] \arrow[r, phantom, "\Downarrow\alpha"] & \mathrm{Id}_y \end{tikzcd} \right) \in \mathrm{Map}_{\mathcal{C}}(y, y)_2$$

- A 3-morphism

$$\left(\begin{tikzcd} \mathrm{Id}_y \arrow[r, bend left=60, "\mathrm{Id}_{\mathrm{Id}_y}"] \arrow[r, bend right=60, "\rho\circ\iota"'] \arrow[r, phantom, "\Downarrow\beta"] & \mathrm{Id}_y \end{tikzcd}\right) \in \mathrm{Map}_{\mathcal{C}}(y,\, y)_2$$

- An equiality

$$\begin{tikzcd} \mathrm{Id}_y \arrow[r, bend left=60, "\mathrm{Id}_{\mathrm{Id}_y}"] \arrow[r, "\rho\circ\iota"] \arrow[r, bend right=60, "\mathrm{Id}_{\mathrm{Id}_y}"'] \arrow[r, phantom, shift left=3, "\Downarrow\beta"] \arrow[r, phantom, shift right=3, "\Downarrow\alpha"] & \mathrm{Id}_y \end{tikzcd} = \begin{tikzcd} \mathrm{Id}_y \arrow[r, bend left=60, "\mathrm{Id}_{\mathrm{Id}_y}"] \arrow[r, bend right=60, "\mathrm{Id}_{\mathrm{Id}_y}"'] \arrow[r, phantom, "\Downarrow\mathrm{Id}_{\mathrm{Id}_{\mathrm{Id}_y}}"] & \mathrm{Id}_y \end{tikzcd}$$

【例 5.1.4】4-condensation

Let $\mathcal{C}$ be a (4, 4)-category. By **【例 5.1.2】**, **3-condensation** of x onto y in $\mathcal{C}$ consists of these data:

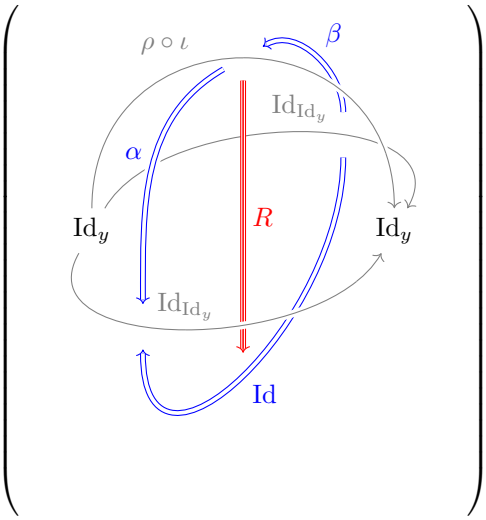
- A 1-morphism $r \in \mathrm{Map}_{\mathcal{C}}(x,\, y)_0$
- A 1-morphism $i \in \mathrm{Map}_{\mathcal{C}}(y,\, x)_0$
- A 2-morphism $(r \circ i \xrightarrow{\rho} \mathrm{Id}_y) \in \mathrm{Map}_{\mathcal{C}}(y,\, y)_1$
- A 2-morphism $(\mathrm{Id}_y \xrightarrow{\iota} r \circ i) \in \mathrm{Map}_{\mathcal{C}}(y,\, y)_1$
- A 3-morphism

$$\left(\begin{tikzcd} \mathrm{Id}_y \arrow[r, bend left=60, "\rho\circ\iota"] \arrow[r, bend right=60, "\mathrm{Id}_{\mathrm{Id}_y}"'] \arrow[r, phantom, "\Downarrow\alpha"] & \mathrm{Id}_y \end{tikzcd}\right) \in \mathrm{Map}_{\mathcal{C}}(y,\, y)_2$$

- A 3-morphism

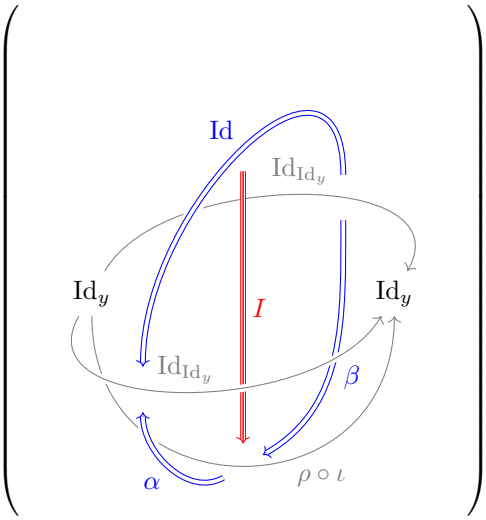
$$\left(\begin{tikzcd} \mathrm{Id}_y \arrow[r, bend left=60, "\mathrm{Id}_{\mathrm{Id}_y}"] \arrow[r, bend right=60, "\rho\circ\iota"'] \arrow[r, phantom, "\Downarrow\beta"] & \mathrm{Id}_y \end{tikzcd}\right) \in \mathrm{Map}_{\mathcal{C}}(y,\, y)_2$$

- A 4-morphism



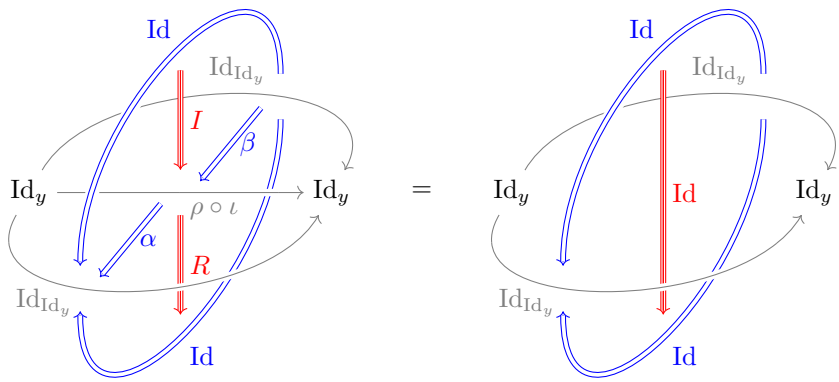
$\in \mathrm{Map}_{\mathcal{C}}(y,\, y)_3$

- A 4-morphism



$\in \mathrm{Map}_{\mathcal{C}}(y,\, y)_3$

- An equality



5.1.1 Walking condensation

Let $\mathcal{C}$ be an $(n,\, n)$-category. Roughly speaking, a **walking n-condensation** is a (strict?) n-category $\spadesuit_n$ which "generates" n-condensation in $\mathcal{C}$. i.e. the functor category $\mathcal{F}\mathbf{un}(\spadesuit_n,\, \mathcal{C})$ is equivalent to the category of n-condensations in $\mathcal{C}$.

5.1.2 Condensation algebra / monad

Fix an $(n,\, n)$-category $\mathcal{C}$. Now we will outline the definition of **condensation monad in $\mathcal{C}$**.

Let's pick out an object $x \in \mathcal{C}_0$. Recall that the endmorphism $(n-1,\, n-1)$-category $\mathrm{End}_{\mathcal{C}}(x) \coloneqq \mathrm{Map}_{\mathcal{C}}(x,\, x)$ has a natural monoidal structure induced by a composition of 1-morphisms in $\mathcal{C}$. Let $\clubsuit_n \subset \spadesuit_n$ be a subcategory [*2] which consists of walking n-condensations with a single object. A **condensation monad** in $\mathcal{C}$ is a functor $(x \in \mathcal{C}_0;\, e \in \mathrm{End}_{\mathcal{C}}(x)_0,\, e^{\circ 2} \twoheadrightarrow e,\, \dots)\colon \clubsuit_n \longrightarrow \mathcal{C}$. An $(n,\, n)$-category $\mathcal{C}$ is said to **have all condensates** if each condensation monad $\clubsuit_n \xrightarrow{A} \mathcal{C}$ extends to an n-condensation $\spadesuit_n \xrightarrow{\bar{A}} \mathcal{C}$:

$$\begin{tikzcd} \clubsuit_n \arrow[r, "A"] \arrow[d, hook] & \mathcal{C} \\ \spadesuit_n \arrow[ur, dashed, "\bar{A}"'] & \end{tikzcd}$$

However, there are some technical issues in this definition , so we should work on more concrete definition.

[*2] (18 February, 2026) Full-subness of $\clubsuit_n$ is still a conjecture [?].

定義 5.3: condensation algebra / monad

Let $n \geq 0$ and let $\mathcal{V}^{\otimes}$ be a <u>monoidal</u> (n, n)-category [a]. A **condensation algebra** in $\mathcal{V}^{\otimes}$ is a sequence of "commuting condensation cubes":

$$A \in (\mathcal{V}^{\otimes})_0, \quad A \otimes A \twoheadrightarrow A, \qquad \begin{tikzcd} A^{\otimes 3} \arrow[r, two heads] \arrow[d, two heads] & A^{\otimes 2} \arrow[d, two heads] \\ A^{\otimes 2} \arrow[r, two heads] & A \end{tikzcd}, \qquad \begin{tikzcd} A^{\otimes 4} \arrow[rr, two heads] \arrow[dr, two heads] \arrow[dd, two heads] & & A^{\otimes 3} \arrow[dr, two heads] \arrow[dd, two heads] & \\ & A^{\otimes 3} \arrow[rr, two heads] \arrow[dd, two heads] & & A^{\otimes 2} \arrow[dd, two heads] \\ A^{\otimes 3} \arrow[rr, two heads] \arrow[dr, two heads] & & A^{\otimes 2} \arrow[dr, two heads] & \\ & A^{\otimes 2} \arrow[rr, two heads] & & A \end{tikzcd}, \quad \cdots$$

in which the faces of the k-dimensional cube are either whiskerings of lower-dimensional cubes or are identities [**?**, Definition 2.2.1.].

[a] **Monoidal (0, 0)-category** is a set $\mathcal{V}_0$ with a map $\otimes \colon \mathcal{V}_0 \times \mathcal{V}_0 \longrightarrow \mathcal{V}_0$ and a distinguished element $1 \in \mathcal{V}_0$, which makes up of a monoid in $\mathbf{Sets} = \mathbf{Cat}_{(0,\,0)}$.

Let $\mathcal{C}$ be an (n, n)-category[a]. A **condensation monad** in $\mathcal{C}$ is a pair of an object $x \in \mathcal{C}_0$ and a condensation algebra in the monoidal $(n-1, n-1)$-category $\mathrm{End}_{\mathcal{C}}(x)$.

[a] Not necessarily monoidal.

We would like to construct a **higher Morita category of condensation algebras** in $\mathcal{V}^{\otimes}$.

定義 5.4: condensation bimodule

Let $n \geq 0$ and let $\mathcal{V}^{\otimes}$ be a <u>monoidal</u> (n, n)-category. Pick two condensation algebras $(A, A^{\otimes 2} \twoheadrightarrow A, \dots)$, $(B, B^{\otimes 2} \twoheadrightarrow B, \dots)$ in $\mathcal{V}^{\otimes}$.

A **condensation (A, B)-bimodule** in $\mathcal{V}^{\otimes}$ is a sequence of "commuting condensation cubes" [**?**, Definition 2.3.3.]:

$$M \in (\mathcal{V}^{\otimes})_0, \qquad A \otimes M \twoheadrightarrow M, \qquad M \otimes B \twoheadrightarrow M,$$

$$\begin{tikzcd} A^{\otimes 2} \otimes M \arrow[r, two heads] \arrow[d, two heads] & A \otimes M \arrow[d, two heads] \\ A \otimes M \arrow[r, two heads] & M \end{tikzcd}, \qquad \begin{tikzcd} A \otimes M \otimes B \arrow[r, two heads] \arrow[d, two heads] & M \otimes B \arrow[d, two heads] \\ A \otimes M \arrow[r, two heads] & M \end{tikzcd}, \qquad \begin{tikzcd} M \otimes B^{\otimes 2} \arrow[r, two heads] \arrow[d, two heads] & M \otimes B \arrow[d, two heads] \\ M \otimes B \arrow[r, two heads] & M \end{tikzcd}, \quad \cdots$$

in which $\forall i, j \in \mathbb{Z}_{\geq 0}$, there is a $i+j$-dimensional cube which has $A^{\otimes i} \otimes M \otimes B^{\otimes j} \in (\mathcal{V}^{\otimes})_0$ on its "top" vertex and $M \in (\mathcal{V}^{\otimes})_0$ on its "bottom" vertex.

Let $\mathcal{C}$ be an $(n,\, n)$-category[a] and pick two condensation monads:

- A condensation monad $(x \in \mathcal{C}_0;\, e_1,\, e_1^{\circ 2} \twoheadrightarrow e_1,\, \dots)$ in $\mathrm{End}_{\mathcal{C}}(x)$
- A condensation monad $(y \in \mathcal{C}_0;\, e_2,\, e_2^{\circ 2} \twoheadrightarrow e_2,\, \dots)$ in $\mathrm{End}_{\mathcal{C}}(y)$

A **condensation** $(e_1,\, e_2)$**-bimodule** is a sequence of "commuting condensation cubes" in the $(n-1,\, n-1)$-category $\mathrm{Map}_{\mathcal{C}}(y,\, x)$ [**?**, Definition 2.3.3.]:

$$m \in \mathrm{Map}_{\mathcal{C}}(y,\, x)_0, \qquad e_1 \circ m \twoheadrightarrow m, \qquad m \circ e_2 \twoheadrightarrow m,$$

$$\begin{tikzcd} e_1^{\circ 2} \circ m \arrow[r, two heads] \arrow[d, two heads] & e_1 \circ m \arrow[d, two heads] \\ e_1 \circ m \arrow[r, two heads] & m \end{tikzcd}, \quad \begin{tikzcd} e_1 \circ m \circ e_2 \arrow[r, two heads] \arrow[d, two heads] & m \circ e_2 \arrow[d, two heads] \\ e_1 \circ m \arrow[r, two heads] & m \end{tikzcd}, \quad \begin{tikzcd} m \circ e_2^{\circ 2} \arrow[r, two heads] \arrow[d, two heads] & m \circ e_2 \arrow[d, two heads] \\ m \circ e_2 \arrow[r, two heads] & m \end{tikzcd}, \quad \cdots$$

in which $\forall i,\, j \in \mathbb{Z}_{\geq 0}$, there is a $i+j$-dimensional cube which has $e_1^{\circ i} \circ m \circ e_2^{\circ j} \in \mathrm{Map}_{\mathcal{C}}(y,\, x)_0$ on its "top" vertex and $m \in \mathrm{Map}_{\mathcal{C}}(y,\, x)_0$ on its "bottom" vertex.

[a] Not necessarily monoidal.

5.2 Physical interpretation

5.3 The $(n+1)$-category of linear n-categories

5.3.1 The $(n+1)$-category of presentable n-categories

Let $\mathbf{Pr}^{\mathrm{L}}$ be the symmetric monoidal $(\infty,\, 2)$-category of presentable $(\infty,\, 1)$-categories, colimit-preserving functors, and natural transformations [**?**, Definition 5.5.3.1.]. After [**?**, Definition 1.2.] [**?**, DEFINITION 1.15], we define the $(\infty,\, n+1)$-category $n\mathbf{Pr}$ of presentable $(\infty,\, n)$-categories as follows:

定義 5.5:

Let $1\mathbf{Pr} \coloneqq \mathbf{Pr}^{\mathrm{L}}$. For each $n \geq 2$, the symmetric monoidal "$(\infty,\, n+1)$-category"[a] of presentable $(\infty,\, n)$-categories is defined inductively as

$$n\mathbf{Pr} \coloneqq \mathbf{RMod}_{(n-1)\mathbf{Pr}}(1\mathbf{Pr}).$$

[a] Strictly speaking, our definition presents $n\mathbf{Pr}$ as an $(\infty,\, 1)$-category. However, in some sense, it can also be regarded as an $(\infty,\, n+1)$-category [**?**, REMARK 1.16.].

5.3.2 Enrichment vs. module structure

The following is a very rough and intuitive discussion.

We can regard an arbitrary object $\mathcal{C} \in \mathrm{Ob}(n\mathbf{Pr})$ as an "$(\infty, 1)$-*category* $\mathcal{C} \in \mathrm{Ob}(1\mathbf{Pr})$ *enriched by the symmetric monoidal* (∞, n)-*category* $(n-1)\mathbf{Pr}$".

By definition we have a module action functor

$$\blacktriangleleft \colon \mathcal{C} \times (n-1)\mathbf{Pr} \longrightarrow \mathcal{C}.$$

The representability of the functor (i.e. the existence of the internal hom) [*3]

$$\begin{aligned} \mathrm{Map}_{\mathcal{C}}\big(x \blacktriangleleft \text{-}, y\big) \colon (n-1)\mathbf{Pr} &\longrightarrow \mathcal{S}\mathbf{paces}, \\ \mathcal{V} &\longmapsto \mathrm{Map}_{\mathcal{C}}\big(x \blacktriangleleft \mathcal{V}, y\big) \end{aligned}$$

for each $x, y \in \mathrm{Ob}(\mathcal{C})$ guarantees the equivalence of mapping spaces

$$\mathrm{Map}_{\mathcal{C}}(x \blacktriangleleft \mathcal{V}, y) \simeq \mathrm{Map}_{(n-1)\mathbf{Pr}}\big(\mathcal{V}, \underline{\mathrm{Map}_{\mathcal{C}}}(x, y)\big) \tag{5.3.1}$$

for an arbitrary presentable $(n-1)$-categories $\mathcal{V} \in \mathrm{Ob}((n-1)\mathbf{Pr})$. We then regard the equivalence (5.3.1) as the definition of the "*mapping* $(\infty, n-1)$-*category*" $\underline{\mathrm{Map}_{\mathcal{C}}}(x, y) \in \mathrm{Ob}((n-1)\mathbf{Pr})$.

5.3.3 The $(n+1)$-category of $\mathbb{K}$-linear presentable n-categories

A similar consideration leads us to the definition of "$\mathbb{K}$-linear presentable (∞, n)-categories". Underlying intuition is "the presentable (∞, n)-category enriched by $\mathbb{K}$-linear presentable $(\infty, n-1)$-categories".

定義 5.6: [?, DEFINITION 1.17.]

Let $\mathbb{K}$ be an algebraically closed field. Let $0\mathbf{Pr}_{\mathbb{K}} \coloneqq \mathbf{Vec}_{\mathbb{K}}$.

Then, for each $n \geq 1$, the symmetric monoidal "$(\infty, n+1)$-category" of $\mathbb{K}$-linear presentable (∞, n)-categories is defined inductively as

$$n\mathbf{Pr}_{\mathbb{K}} \coloneqq \mathbf{RMod}_{(n-1)\mathbf{Pr}_{\mathbb{K}}}(n\mathbf{Pr})$$

【例 5.3.1】 The 2-category of 2-vector spaces $1\mathbf{Pr}_{\mathbb{K}}$

Assume $n = 1$. Roughly speaking, we can regard $1\mathbf{Pr}$ as the bicategory of $(1, 1)$-categories, functors, and natural transformations. For each $\mathcal{C} \in \mathrm{Ob}(1\mathbf{Pr}_{\mathbb{K}})$, we have a module action functor

$$\blacktriangleleft \colon \mathcal{C} \times \mathbf{Vec}_{\mathbb{K}} \longrightarrow \mathcal{C}.$$

If the functor

$$\mathrm{Hom}_{\mathcal{C}}(x \blacktriangleleft \text{-}, y) \colon \mathbf{Vec}_{\mathbb{K}} \longrightarrow \mathbf{Sets},$$

[*3] $\mathcal{S}\mathbf{paces}$ is the $(\infty, 1)$-category of Kan complexes (i.e. $(\infty, 0)$-categories). Recently some higher categorists use the terminology $\mathcal{A}\mathbf{ni}$ ($(\infty, 1)$-category of *animas*) instead of $\mathcal{S}\mathbf{paces}$.

$$V \longmapsto \mathrm{Hom}_{\mathcal{C}}(x \blacktriangleleft V,\, y)$$

is representable for each $x,\, y \in \mathrm{Ob}(\mathcal{C})$, we have a bijection of 0-categories (i.e. sets)

$$\mathrm{Hom}_{\mathcal{C}}(x \blacktriangleleft V,\, y) \simeq \mathrm{Hom}_{\mathbf{Vec}_{\mathbb{K}}}\big(V,\, \underline{\mathrm{Hom}_{\mathcal{C}}}(x,\, y)\big)$$

for an arbitrary vector space $V \in \mathrm{Ob}(\mathbf{Vec}_{\mathbb{K}})$. Such a bijection guarantees the existence of mapping vector space $\underline{\mathrm{Hom}_{\mathcal{C}}}(x,\, y) \in \mathrm{Ob}(\mathbf{Vec}_{\mathbb{K}})$, and therefore recovers $\mathcal{C}$ as an ordinary $\mathbf{Vec}_{\mathbb{K}}$-enriched category.

Note that $\vec{\mathbb{K}} \in \mathrm{Ob}(1\mathbf{Pr}_{\mathbb{K}})$. The module action functor of $\mathbf{Vec}_{\mathbb{K}} = 0\mathbf{Pr}_{\mathbb{K}}$ is the ordinary tensor product

$$\otimes_{\mathbb{K}} \colon \mathbf{Vec}_{\mathbb{K}} \times \mathbf{Vec}_{\mathbb{K}} \longrightarrow \mathbf{Vec}_{\mathbb{K}}$$

of vector spaces.

【例 5.3.2】The 3-category of 3-vector spaces $2\mathbf{Pr}_{\mathbb{K}}$

Assume $n = 2$. Roughly speaking, we can regard $2\mathbf{Pr}$ as the tricategory of bicategories, pseudofunctors, strong transformations, and modifications [**?**]. For each $\mathcal{C} \in \mathrm{Ob}(2\mathbf{Pr}_{\mathbb{K}})$, we have a module action pseudofunctor

$$\blacktriangleleft \colon \mathcal{C} \times 1\mathbf{Pr}_{\mathbb{K}} \longrightarrow \mathcal{C}.$$

If the pseudofunctor

$$\begin{aligned} \mathrm{Map}_{\mathcal{C}}(x \blacktriangleleft \text{-},\, y) \colon 1\mathbf{Pr}_{\mathbb{K}} &\longrightarrow \mathbf{Cat}, \\ \mathcal{V} &\longmapsto \mathrm{Map}_{\mathcal{C}}(x \blacktriangleleft \mathcal{V},\, y) \end{aligned}$$

is representable for each $x,\, y \in \mathrm{Ob}(\mathcal{C})$, we have an equivalence of 1-categories

$$\mathrm{Map}_{\mathcal{C}}(x \blacktriangleleft \mathcal{V},\, y) \simeq \mathrm{Map}_{1\mathbf{Pr}_{\mathbb{K}}}\big(\mathcal{V},\, \underline{\mathrm{Map}_{\mathcal{C}}}(x,\, y)\big)$$

for an arbitrary $\mathbb{K}$-linear 1-category $\mathcal{V} \in \mathrm{Ob}(1\mathbf{Pr}_{\mathbb{K}})$. Such an equivalence guarantees the existence of mapping $\mathbb{K}$-linear 1-category $\underline{\mathrm{Map}_{\mathcal{C}}}(x,\, y) \in \mathrm{Ob}(1\mathbf{Pr}_{\mathbb{K}})$. i.e. $\mathcal{C}$ can be regarded as a bicategory whose collection of 2-morphisms constitutes a $\mathbb{K}$-vector space.

Note that $1\mathbf{Pr}_{\mathbb{K}}$ itself is an object of $2\mathbf{Pr}_{\mathbb{K}}$. The module action pseudofunctor of $1\mathbf{Pr}_{\mathbb{K}}$ is the Lurie's tensor product[a]:

$$\boxtimes \colon 1\mathbf{Pr}_{\mathbb{K}} \times 1\mathbf{Pr}_{\mathbb{K}} \longrightarrow 1\mathbf{Pr}_{\mathbb{K}}$$

[a] Analogous to the classical Deligne tensor product.

! From now on, we use the terminology

$$\boldsymbol{n}\mathbf{Vec}_{\mathbb{K}} \coloneqq (n-1)\mathbf{Pr}_{\mathbb{K}} \quad (n \geq 1).$$

An object of $\boldsymbol{n}\mathbf{Vec}_{\mathbb{K}}$ is called an *n-vector space*.

5.4 Tannaka-Krein reconstruction of 3-vector spaces

5.4.1 The choice of the fiber 2-functor

Fix a 3-vector space $\mathcal{C} \in \mathrm{Ob}(\mathbf{3Vec}_{\mathbb{K}})$. Tannaka-Krein reconstruction needs a faithful pseudofunctor $F^{(1)} \in \mathrm{Ob}(\mathrm{Map}_{\mathbf{3Vec}_{\mathbb{K}}}(\mathcal{C}, \mathbf{2Vec}_{\mathbb{K}})^{\mathrm{faith}})$. We denote

$$T_{\mathcal{C}} := \mathrm{End}_{\mathrm{Map}_{\mathbf{3Vec}_{\mathbb{K}}}(\mathcal{C}, \mathbf{2Vec}_{\mathbb{K}})}(F^{(1)}) \in \mathrm{Ob}(\mathbf{2Vec}_{\mathbb{K}}).$$

If $F^{(1)}$ admits a left adjoint $F^{(1)\vee} \in \mathrm{Ob}(\mathrm{Map}_{\mathbf{3Vec}_{\mathbb{K}}}(\mathbf{2Vec}_{\mathbb{K}}, \mathcal{C}))$ (automatic?) , an isomorphism

$$T_{\mathcal{C}} \simeq \mathrm{Map}_{\mathrm{Map}_{\mathbf{3Vec}_{\mathbb{K}}}(\mathbf{2Vec}_{\mathbb{K}}, \mathbf{2Vec}_{\mathbb{K}})}(\mathrm{Id}_{\mathbf{2Vec}_{\mathbb{K}}}, F^{(1)} \circ F^{(1)\vee})$$

in $\mathbf{2Vec}_{\mathbb{K}}$ [*4] exists.

補題 5.1:

There is an isomorphism

$$I \colon \mathrm{Map}_{\mathbf{3Vec}_{\mathbb{K}}}(\mathbf{2Vec}_{\mathbb{K}}, \mathbf{2Vec}_{\mathbb{K}}) \xrightarrow{\simeq} \mathbf{2Vec}_{\mathbb{K}}$$

in $\mathbf{3Vec}_{\mathbb{K}}$.

証明 Probably in [?] ■

Equivalence I in Lemma 5.1 induces an isomorphism[*5]

$$T_{\mathcal{C}} \simeq \mathrm{Map}_{\mathbf{2Vec}_{\mathbb{K}}}\big(\mathbf{Vec}_{\mathbb{K}}, I(F^{(1)} \circ F^{(1)\vee})\big)$$

in $\mathbf{2Vec}_{\mathbb{K}}$.

命題 5.1:

$$T_{\mathcal{C}} \simeq I(F^{(1)} \circ F^{(1)\vee})$$

証明 For each $\mathcal{V} \in \mathrm{Ob}(\mathbf{2Vec}_{\mathbb{K}})$, 【例 5.3.2】shows the natural isomorphism (i.e. a kind of tensor-hom adjunction)

$$\mathrm{Map}_{\mathbf{2Vec}_{\mathbb{K}}}\Big(\mathcal{V}, \mathrm{Map}_{\mathbf{2Vec}_{\mathbb{K}}}\big(\mathbf{Vec}_{\mathbb{K}}, I(F^{(1)} \circ F^{(1)\vee})\big)\Big)$$

[*4] This is merely an analogue of 【例 5.3.2】. We need some references or a proof. Note that $\mathbf{3Vec}_{\mathbb{K}} \in \mathrm{Ob}(\mathbf{4Vec}_{\mathbb{K}})$. Therefore, by the "tensor-hom adjunction", we can regard $\mathrm{Map}_{\mathbf{3Vec}_{\mathbb{K}}}(\mathcal{C}, \mathbf{2Vec}_{\mathbb{K}})$, $\mathrm{Map}_{\mathbf{3Vec}_{\mathbb{K}}}(\mathbf{2Vec}_{\mathbb{K}}, \mathbf{2Vec}_{\mathbb{K}}) \in \mathrm{Ob}(\mathbf{3Vec}_{\mathbb{K}})$. Furthermore, according to 【例 5.3.2】, we can regard $T_{\mathcal{C}} \in \mathrm{Ob}(\mathbf{2Vec}_{\mathbb{K}})$.

[*5] Sice I is a functor, $I(\mathrm{Id}_{\mathbf{2Vec}_{\mathbb{K}}}) \simeq \mathbf{Vec}_{\mathbb{K}}$.

$$\simeq \mathrm{Map}_{\mathbf{2Vec}_{\mathbb{K}}}\big(\mathbf{Vec}_{\mathbb{K}} \boxtimes \mathcal{V},\, I(F^{(1)} \circ F^{(1)\vee})\big)$$
$$\simeq \mathrm{Map}_{\mathbf{2Vec}_{\mathbb{K}}}\big(\mathcal{V},\, I(F^{(1)} \circ F^{(1)\vee})\big)$$

in **Cat**. i.e. there is an equivalence between two pseudofunctors;

$$\mathrm{Map}_{\mathbf{2Vec}_{\mathbb{K}}}\Big(\text{-},\, \mathrm{Map}_{\mathbf{2Vec}_{\mathbb{K}}}\big(\mathbf{Vec}_{\mathbb{K}},\, I(F^{(1)} \circ F^{(1)\vee})\big)\Big) \simeq \mathrm{Map}_{\mathbf{2Vec}_{\mathbb{K}}}\big(\text{-},\, I(F^{(1)} \circ F^{(1)\vee})\big).$$

Then, Yoneda lemma for bicategories [**?**, Lemma 8.3.11.] induces the equivalence

$$\mathrm{Map}_{\mathbf{2Vec}_{\mathbb{K}}}\big(\mathbf{Vec}_{\mathbb{K}},\, I(F^{(1)} \circ F^{(1)\vee})\big) \simeq I(F^{(1)} \circ F^{(1)\vee})$$

■

5.4.2 The second choice

We have another choice of a faithful functor $F^{(2)} \in \mathrm{Ob}(\mathrm{Map}_{\mathbf{2Vec}_{\mathbb{K}}}(T_{\mathcal{C}},\, \mathbf{Vec}_{\mathbb{K}})^{\mathrm{faith}})$. Combining the Lemma 5.1 and the Proposition 5.1, such a choice is equivalent to the choice of 2-morphism

$$\widehat{F}^{(2)} := I^{-1}(F^{(2)})\colon F^{(1)} \circ F^{(1)\vee} \Longrightarrow \mathrm{Id}_{\mathbf{2Vec}_{\mathbb{K}}}$$

in $\mathbf{3Vec}_{\mathbb{K}}$.

5.5 Dualizability in higher Morita categories